FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
<u>OPEN TEST – II</u>
PAPER –2
<u>TEST DATE: 20-03-2022</u>

<u>ANSWERS, HINTS & SOLUTIONS</u>

Physics PART – I

Section – A

1. A, D

Sol. $\frac{\omega h}{\sin\theta} = v\sin\theta$

$\Rightarrow \omega = \frac{v\sin^2\theta}{h}$

At $\theta = 30°$, $\omega = \frac{8}{1}\times\frac{1}{4} = 2$ rad/s

Now, $\frac{d\omega}{dt} = \frac{v}{h}2\sin\theta\cos\theta\frac{d\theta}{dt}$

$\alpha = \frac{v}{h}\sin 2\theta(-\omega)$

$\alpha = -\frac{v\omega\sin 2\theta}{h}$

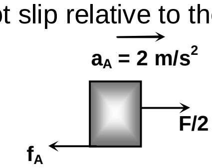
At $\theta = 30°$, $\alpha = \frac{-8\times 2}{1}\times\frac{\sqrt{3}}{2} = -8\sqrt{3}$ rad/s^2

2. A, B, C, D

Sol. The block B will slip relative to the plank C but the block A will not slip relative to the plank C.

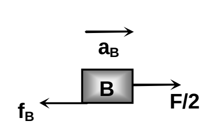
$f_B = 0.1 \times 10 = 1$ N

$f_A + f_B = m_C a_C \Rightarrow f_A + 1 = 1 \times 2 \Rightarrow f_A = 1$ N



$\frac{F}{2} - f_A = m_A a_A \Rightarrow \frac{F}{2} - 1 = 4\times 2$

$\Rightarrow F = 18$ N

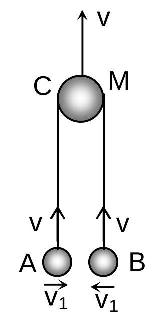


$\frac{F}{2} - f_B = m_B a_B$

$9 - 1 = 1 \times a_0 \Rightarrow a_B = 8$ m/s^2

3. A, B, C

Sol. When the discs A and B are about to collides using conservation of momentum.



$Mv_0 = (M + 2m)v$

$$v = \frac{Mv_0}{M+2m} = \frac{2\times 4}{4} = 2\ \text{m/s} \qquad \ldots(i)$$

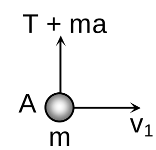
Using conservation of energy,

$$\frac{1}{2}Mv_0^2 = \frac{1}{2}Mv^2 + 2\times\frac{1}{2}m(v^2+v_1^2)$$

$$v_0^2 = 2v^2 + v_1^2$$

$$16 = 8 + v_1^2 \Rightarrow v_1 = 2\sqrt{2}\ \text{m/s} \qquad \ldots(ii)$$

Retardation of the middle disc C,



$$a = \frac{2T}{M} = \frac{2T}{2} = T \qquad \ldots(iii)$$

Now, $T + ma = \dfrac{mv_1^2}{\ell}$

$$T + T = \frac{1\times 8}{1}$$

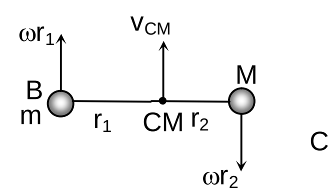
$2T = 8 \Rightarrow T = 4N$

The velocity of the disc A,

$$v_A = \sqrt{v^2+v_1^2} = \sqrt{4+8} = 2\sqrt{3}\ \text{m/s}$$

4. A, D

Sol. The bead and the centre of the ring will move along the circular paths about their centre of mass with a constant angular velocity, $\omega = \dfrac{u}{R} = \dfrac{5}{0.5} = 10$ rad/s



$N = m\omega^2 r = 4 \times 100 \times 0.3 = 120$ N

The velocity of their centre of mass

$$v_{CM} = \frac{mu+0}{m+M} = \frac{4\times 5}{10} = 2\ \text{m/s}$$

The speed of the bead relative to the centre of mass

$$v_1 = \omega r_1 = 10\times 0.3 = 3\ \text{m/s}$$

$$K_{1_{(min)}} = \frac{1}{2}m(v_1 - v_{cm})^2 = \frac{1}{2}\times 4\times(3-2)^2 = 2\ \text{J}$$

5. A, C

Sol. $P_1A = kx_1 \qquad \ldots(i)$

$P_2A = kx_2 \qquad \ldots(ii)$

Now, $P_1A\ell_1 = nRT_1 \Rightarrow kx_1 = \dfrac{RT_1}{\ell_1} \qquad \ldots(iii)$

$P_2A\ell_2 = nRT_2 \Rightarrow kx_2 = \dfrac{RT_2}{\eta\ell_1} \qquad \ldots(iv)$

Adding equation (iii) and (iv), we get

$$k(x_2 + x_1) = \frac{R}{\eta\ell_1}(\eta T_1 + T_2)$$

The work done by the gas, $\Delta W = \dfrac{1}{2}k(x_2^2 - x_1^2)$

$$\Rightarrow \Delta W = \frac{k}{2}(x_2 + x_1)(x_2 - x_1)$$

$$\Delta W = \frac{R}{2}\left(\frac{\eta T_1 + T_2}{\eta \ell_1}\right)(\eta - 1)\ell_1$$

$$\Delta W = \frac{R}{2}\left(\frac{\eta - 1}{\eta}\right)(\eta T_1 + T_2) = 605.9 \text{ J}$$

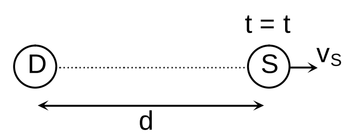
$$\Delta U = nC_V\Delta T = 1 \times \frac{3R}{2}(400 - 300)$$

$\Delta U = 150R = 1245$ J

Hence, $\Delta Q = \Delta U + \Delta W = 1245 + 605.9 = 1850.9$ J

6. A, D

Sol. Le the sound emitted by the source at time 't' is received by the stationary observer at t = 10 sec.



$$d = \frac{1}{2}at^2 = \frac{1}{2} \times 10t^2 = 5t^2$$

Now, d = 340(10 – t)

$5t^2 = 340(10 - t)$

$t^2 + 68t - 680 = 0$

$$t = \frac{-68 \pm \sqrt{(68)^2 + 4 \times 680}}{2}$$

t = 8.8 sec (approximately)

$v_S = at = 10 \times 8.8 = 88$ m/s

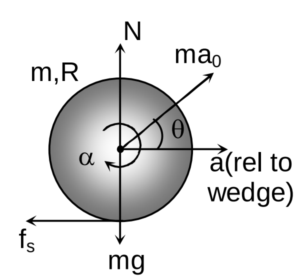
Hence, $f = \left(\frac{v}{v + v_S}\right)f_0$

$$f = \left(\frac{340}{340 + 88}\right)1700 = 1350 \text{ Hz (approximately)}$$

7. C

8. B

Sol. **(for Q. 7-8)**



$N + ma_0 \sin\theta = mg$ …(i)

$ma_0\cos\theta - f_s = ma$ …(ii)

$f_sR = mR^2\alpha$

$f_s = ma$ …(iii)

$(N + Mg)\sin\theta - f_s\cos\theta = Ma_0$ …(iv)

Solving (ii) and (iii), we get

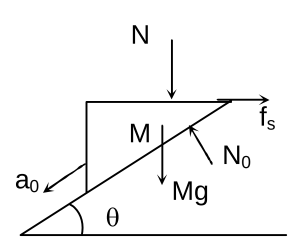
$ma_0\cos\theta = 2ma$

Solving (ii) and (iii) we get

$ma_0\cos\theta = 2ma$

$$a = \frac{a_0\cos\theta}{2} \quad \text{…(v)}$$

Solving (i), (iii) and (iv), we get



$$a_0 = \frac{3g}{5} = 6 \text{ m/s}^2$$

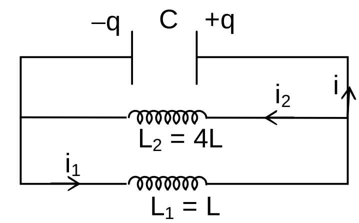
$$a = \frac{a_0\cos\theta}{2} = \frac{3\sqrt{3}}{2} \text{ m/s}^2$$

9. B

10. C

Sol. **(for Q. 9-10)**

After the switch 'S' is shifted to position-2,



$L_1\frac{di_1}{dt} + L_2\frac{di_2}{dt} = 0$

$Ldi_1 + 4Ldi_2 = 0$

$di_1 + 4di_2 = 0$

$\int_{i_0}^{i_1} di_1 = -4\int_0^{i_2} di_2$

$i_1 - i_0 = -4i_2 \Rightarrow i_1 + 4i_2 = i_0$ …(i)

When the charge on the capacitor is maximum, i = 0

Hence $i_1 = i_2$ …(ii)

From (i) and (ii), we get

$i_1 = i_2 = \frac{i_0}{5} = \frac{\varepsilon}{5R}$ $\left(i_0 = \frac{\varepsilon}{R}\right)$

Now, $\frac{q_{max}^2}{2C} = \frac{1}{2}Li_0^2 - \frac{1}{2}(L + 4L)\left(\frac{i_0}{5}\right)^2$

$\frac{q_{max}^2}{2C} = \frac{2LI_0^2}{5} \Rightarrow q_{max} = \frac{\varepsilon}{R}\sqrt{\frac{4CL}{5}}$

When the current through the inductor L_2 is maximum, q = 0

$\frac{1}{2}Li_1^2 + \frac{1}{2}4Li_2^2 = \frac{1}{2}Li_0^2 \Rightarrow i_1^2 + 4i_2^2 = i_0^2$

$\Rightarrow (i_0 + i_1)(i_0 - i_1) = 4i_2^2$

$\Rightarrow (i_0 + i_1)4i_2 = 4i_2^2 \Rightarrow i_1 + i_0 = i_2$

$\Rightarrow i_2 - i_1 = i_0$ …(iii)

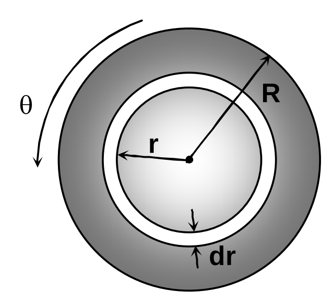
Solving (i) and (iii) we get

$5i_2 = 2i_0 \Rightarrow i_2 = \frac{2i_0}{5} = \frac{2\varepsilon}{5R}$

Section – B

11. 8

Sol. $dU = \frac{1}{2}\eta\left(\frac{r\theta}{\ell}\right)^2 2\pi r dr \ell$



$\int dU = \frac{\pi\eta\theta^2}{\ell}\int_0^R r^3 dr$

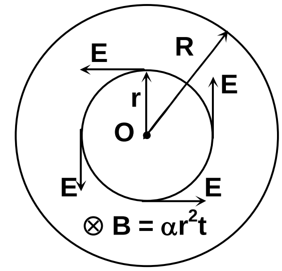
$U = \frac{\pi\eta R^4\theta^2}{4\ell}$

The elastic deformation energy stored in the rod is

$U = \frac{\pi\eta R^4\theta^2}{4\ell}$

12. 4

Sol. When $r \le R$



$$E2\pi r = \frac{d}{dt}\left[\int_0^r B2\pi r dr\right]$$

$$Er = \frac{d}{dt}\left[\int_0^r \alpha r^2 t r dr\right]$$

$$E = \frac{\alpha r^3}{4}$$

When $r > R$

$$E2\pi r = \frac{d}{dt}\left[\int_0^R \alpha r^2 t 2\pi r dr\right]$$

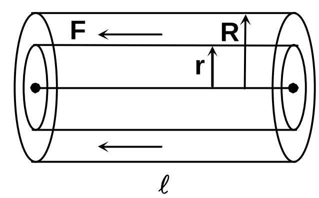
$$E = \frac{\alpha R^4}{4r}$$

At $r = \frac{R}{2}$, $E_1 = \frac{\alpha R^3}{32}$ and at r = 2R, $E_2 = \frac{\alpha R^3}{8}$

Hence, $\frac{E_2}{E_1} = \frac{\alpha R^3/8}{\alpha R^3/32} = 4$

13. 24

Sol. Let the pressure difference at the two ends of the tube is ΔP. The viscous force acting on the cylindrical volume of the liquid of radius 'r' is



$$F = -\eta 2\pi r \ell \frac{dv}{dr}$$

$$F = -\eta 2\pi r \ell v_0 \left(-\frac{2r}{R^2}\right)$$

$$F = 4\pi\eta \ell v_0 \left(\frac{r^2}{R^2}\right) \quad \ldots(i)$$

For the steady flow of the liquid

$\Delta P \pi r^2 = F$

$$\Delta P \pi r^2 = 4\pi\eta \ell v_0 \left(\frac{r^2}{R^2}\right)$$

$$\Delta P = \frac{4\eta \ell v_0}{R^2}$$

Hence, k = 24

Section – C

14. 12.50

15. 2.40

Sol. **(for Q. 14-15)**

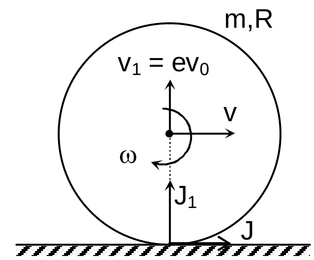
$v_1 = ev_0 = 0.5v_0$

$J_1 = m(v_1 + v_0) = 1.5mv_0$ …(i)

$J_{max} = \mu\int N dt = \mu J_1 = 0.4 \times 1.5mv_0$

$J_{max} = 0.6mv_0$ …(ii)

using conservation of angular momentum about the contact point,



$v = \omega R$

$$\frac{2}{5}mR^2\omega_0 + mv_0R = \frac{2}{5}mR^2\omega + mvR$$

$$\Rightarrow \frac{3}{5}mv_0R + mv_0R = \frac{2}{5}mvR + mvR$$

$$\Rightarrow \frac{8}{5}mv_0R = \frac{7}{5}mvR$$

$$\Rightarrow v = \frac{8v_0}{7}$$

Hence, $\omega = \frac{v}{R} = \frac{8v_0}{7R} = \frac{8\times3.5}{7\times0.32} = 12.50$ rad/s

The impulse due to friction on the sphere

$$J = m(v - v_0) = m\left(\frac{8v_0}{7} - v_0\right) = \frac{mv_0}{7}$$

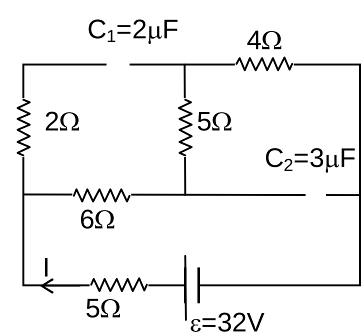
$$J = \frac{mv_0}{7} = \frac{4.8\times3.5}{7} = 2.40 \text{ kg-m/s}$$

16. 35.20

17. 43.20

Sol. **(for Q. 16-17)**

The equivalent circuit is shown in the steady state.



$I = \frac{32}{20} = 1.6$ A

Potential drop across the capacitor C_1

$V_1 = I(6+5) = 1.6\times11 = 17.6$ volt

$q_1 = C_1V_1 = 2\times17.6 = 35.2\ \mu C$

Potential drop across the capacitor C_2,

$V_2 = I(5+4) = 1.6\times9 = 14.4$ volt

$q_2 = C_2V_2 = 3\times14.4 = 43.2\ \mu C$

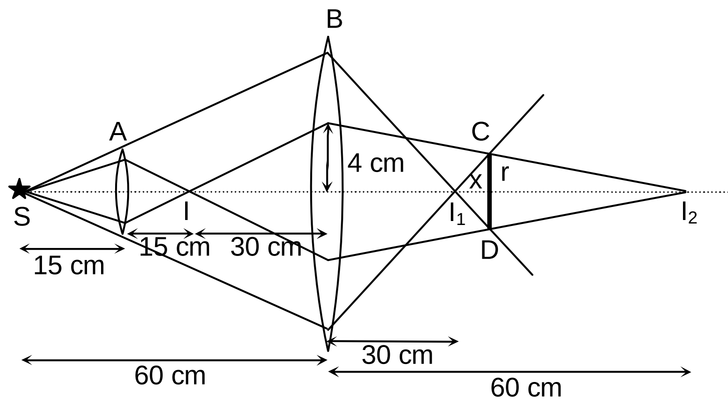
18. 35.00

19. 1.67

Sol. **(for Q. 18-19)**

The position of image I_1 formed due to the direct refraction through the lens B,

$$\frac{1}{v_1} - \frac{1}{(-60)} = \frac{1}{20} \Rightarrow \frac{1}{v_1} = \frac{1}{20} - \frac{1}{60} = \frac{3-1}{60} \Rightarrow v_1 = 30 \text{ cm}$$



The position of image I_2 formed due to double refraction through the lenses A and B,

$\frac{1}{v_2} - \frac{1}{(-30)} = \frac{1}{20} \Rightarrow \frac{1}{v_2} - \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60} \Rightarrow v_2 = 60$ cm

Now, $\frac{r}{x} = \frac{10}{30} \Rightarrow x = 3r$

Also, $\frac{r}{(30-x)} = \frac{4}{60} \Rightarrow 15r = 30 - 3r$

$\Rightarrow r = \frac{30}{18} = \frac{5}{3} = 1.67$ cm

Hence, x = 3r = 5 cm

d = 30 + x = 30 + 5 = 35 cm

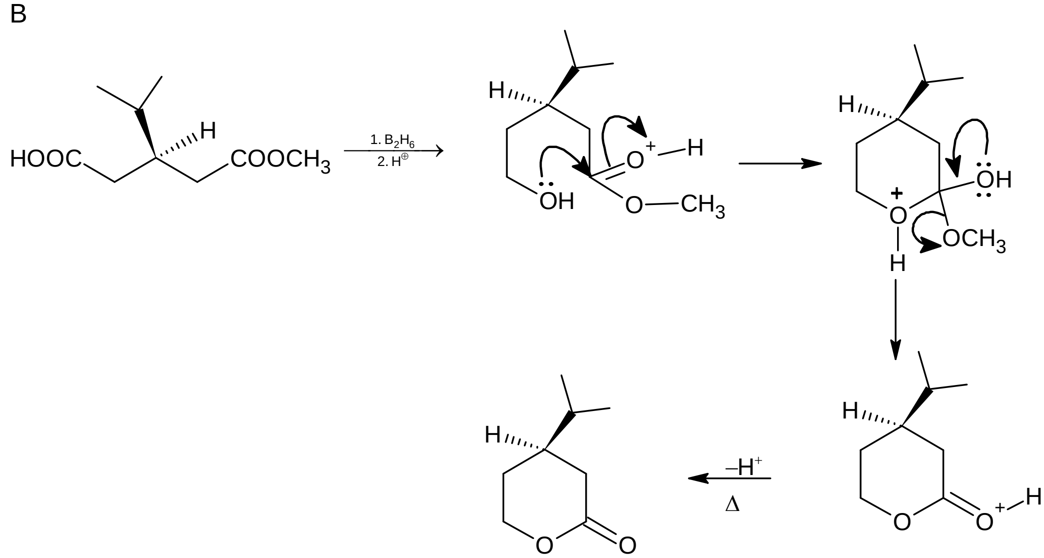
Chemistry

PART – II

Section – A

20. B

Sol.



21. A, C, D

Sol. (A) Because $T_C < T_i$ (inversion),
So, gas shows cooling effect

(B) $\Delta H_{400} + \Delta_{rxn} C_p (400 - 200)$ of $\Delta H_{400} = \Delta H_{200}$

$\Rightarrow \Delta_{rxn} C_P = 0$

$$\Rightarrow \Delta S_{300} = \Delta S_{500} + \Delta_{rxn} C_P \, ln \frac{300}{500}$$

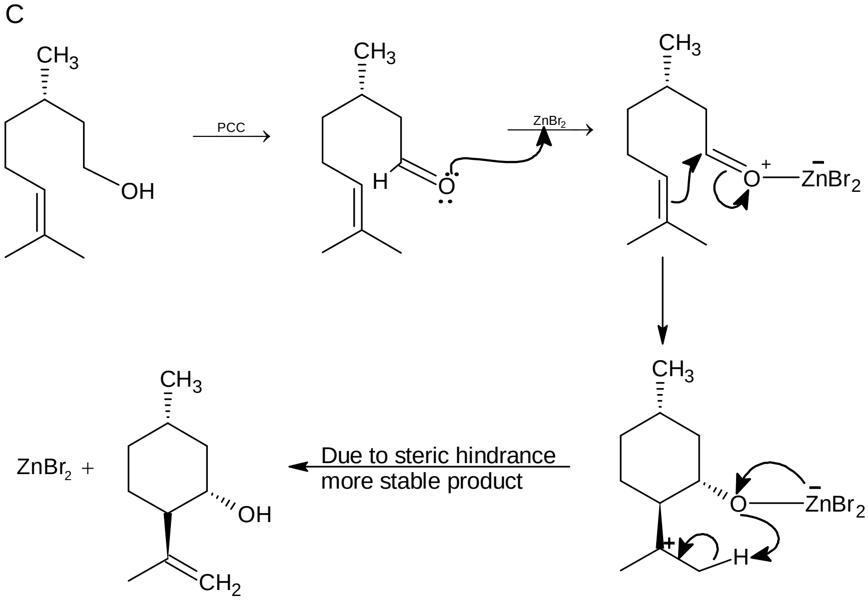
$\Delta S_{300} = \Delta S_{500}$

(C) Z = 1 at Boyle's temperature for limited range of pressure

(D) Equivalent mass of metal chloride is 42.5.

22. C

Sol.



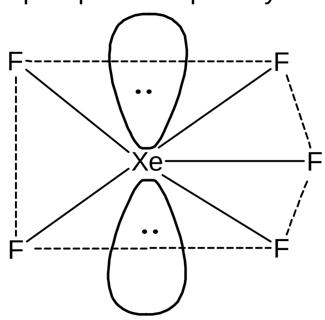
23. B, C, D

Sol. The great thermal and chemical stability of silicones is attributed to high strength of Si - C and (Si – O) bond.

24. A, B, C

Sol. $XeF_5^{-} \to \ell p = 2, bp = 5$

$\ell p + bp = 7 \to sp^3d^3$ hybridized

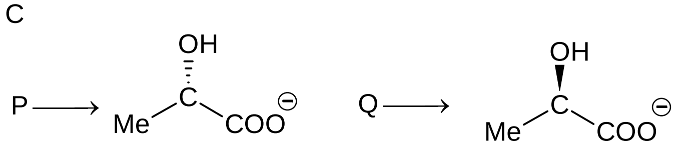


Pentagonal planar shape

$SF_5^-, [IF_5 \text{ and } XeOF_4] \to sp^3d^2$ hybridized $\to$ square pyramidal shape

25. C

Sol.



26. C

Sol. Rate of hydrolysis of II is more than I and III due to steric hindrance. Rate of hydrolysis of III is more than I due to poor back bonding.

27. C

Sol. In compound I, III and V nucleophilic substitution reaction not possible due to double bond character of C – Lg and its option IIIrd due to steric hindrance back attack is not possible.

28. C

29. C

Sol. **(for Q. 28 and 29)**

Since 1/4 th mole will be converted

$\therefore$ in vapours 5 moles total will be present. Assuming x moles to be of X

Minimum pressure for the existence of liquid solution is dew point pressure

$$\frac{1}{P_T} = \left(\frac{Y_x}{P_x^o} + \frac{Y_y}{P_y^o}\right)$$

$P_T = 0.72$ atm

	Liquid	Vapour
X	$10 - x$	x
Y	$5 + x$	$5 - x$

$$\frac{x}{5} = \frac{10 - x}{15} \times \frac{0.6}{P_T} \quad \ldots(1)$$

$$\frac{5 - x}{5} = \frac{5 + x}{15} \times \frac{0.9}{P_T} \quad \ldots(2)$$

Equating P_T from both the equation.

$x^2 + 45x - 100 = 0$

$x = -\frac{45 + \sqrt{2425}}{2} = 2.125$

Mole of X in liquid are = 10 – x
= 7.875

Section – B

30. 144

Sol. $[H^+] = [OH^-] = \frac{it}{96500} = \frac{1.25 \times 212 \times 60}{96500} = 0.165$

At anode

HPO_4^{2-} + H^+ $\rightleftharpoons$ $H_2PO_4^-$

0.330 M 0.165 M 0.330 M

0.330 M – 0.165 M 0 0.330 M + 0.165 M

0.165 M 0.495 M

$$pH = 7.2 + \log\frac{[HPO_4^{2-}]}{[H_2PO_4^-]} = 7.2 + \log\frac{0.165}{0.495}$$

= 7.2 – log 3 = 7.20 – 0.48 = 6.72

At cathode

$H_2PO_4^-$ + OH^- $\rightleftharpoons$ $HPO_4^{--} + H_2O$

0.330 0.165 0.330

0.330 M – 0.165 M 0 0.330 M + 0.165 M

$pH = 7.2 + \log\frac{0.495}{0.165}$

pH = 7.2 + log 3 = 7.20 + 0.48

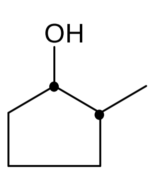
pH = 7.68

= 6.72, y = 7.68

x + y = 14.40

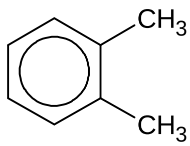
10(x + y) = 144

31. 4

Sol. Product X is  which has 4 streo isomers.

32. 14

Sol. Both the product X and Y are same, i.e. orthoxylene.



Which has DU - 4
Position isomers - 3

Total sum = 4 + 4 + 3 + 3 = 14

Section – C

33. 0.13

34. 53.33

Sol. **(for Q. 33 and 34)**

Pressure of H_2O = 20 torr pressure due to CO, Cl_2 , and $COCl_2$ in final state = 80 tor

$COCl_2 \longrightarrow CO + Cl_2$

100 – x x x

Total pressure = (100 + x) × 2/ 3 = 80

x = 20 torr.)

final (P) $COCl_2$ = 80 × 2/ 3 = 160/3=53.33 torr

Final (P of CO) = final(P of Cl_2) = 20× 2/ 3 = 40/3 torr

Final P (H_2O) = 20 torr ;

Mole fraction of CO = 40/300 = 0.13.

35. 34.65

36. 1.73

Sol. **(for Q. 35 and 36)**

$$\begin{array}{llll} & 2X & +\ Y & \xrightarrow{k}\ P \\ t = 0 \text{ mole} & 2 & 1 & 0 \\ t = 100 \text{ sec} & 2-2x & 1-x & 0.5 \end{array}$$

$$-\frac{1}{2}\frac{d[X]}{dt} = -\frac{d[Y]}{dt} = +\frac{d[P]}{dt} = k[X]$$

Half-life = 100 s [reaction is first order w.r.t. X]

$$k = \frac{0.693}{2\times t_{1/2}} = \frac{0.693}{2\times 100}\sec^{-1} = \frac{0.693\times 10^{-3}}{2} = 34.65\times 10^{-4}\,s^{-1}$$

$$-\frac{d[y]}{dt} = 3.46\times 10^{-3}[X]_{200\text{ s}}$$

200 s means $t_{75\%}$ (for first order $t_{75\%} = 2t_{50\%}$)

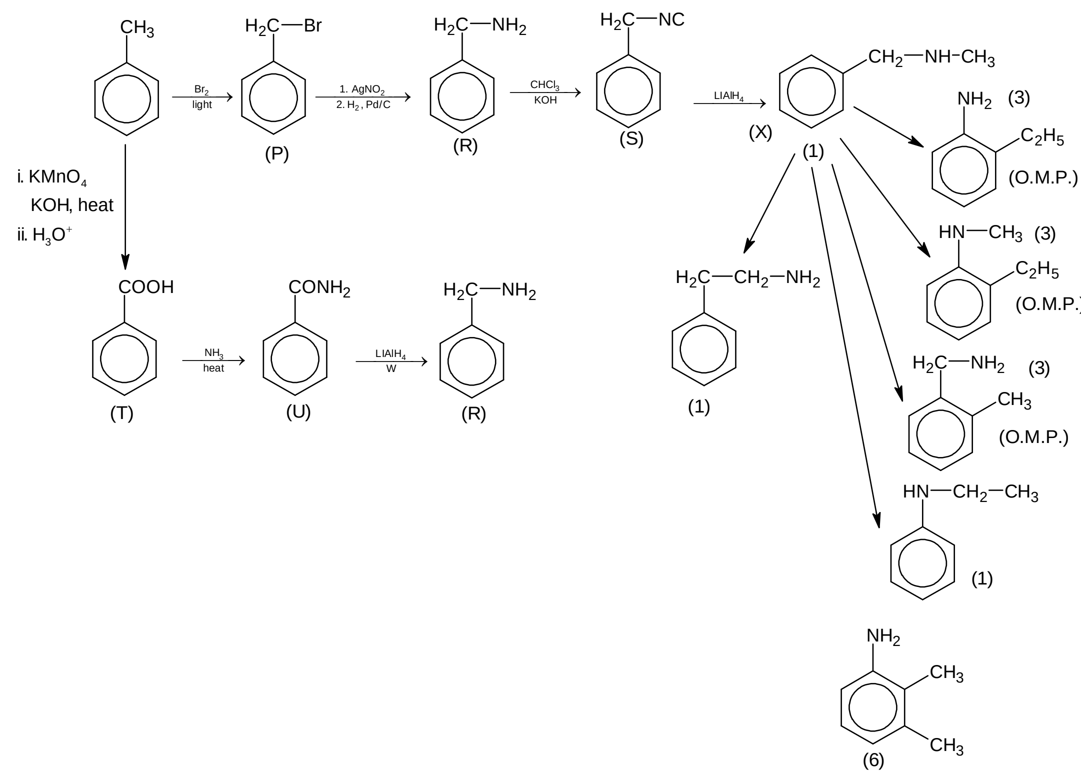
$$\begin{array}{llll} & 2X & +\ Y & \xrightarrow{k}\ P \\ \text{at} = 200 \text{ s} & 0.5 & 0.25 & 0.75 \end{array}$$

$$-\frac{d[Y]}{dt} = 3.46\times 10^{-3}\times(0.5) = 1.73\times 10^{-3}\text{ mol L s}^{-1}$$

37. 394.60

38. 18.00

Sol. Q = $AgNO_2$, R = phenylmethanamine, W = $LiAlH_4$, V = AgCN.



Total isomers = 1 + 1 + 3 + 3 + 3 + 1 + 6 = 18.

Mathematics PART – III

Section – A

39. C, D

Sol. $f(f(f(x))) = 2 - f(x)$

Let f(x) is increasing function $\Rightarrow f(f(f(x)))$ is increasing function for 0 < 2

$\Rightarrow f(f(f(0))) < f(f(f(x))) \Rightarrow 2 - f(0) < 2 - f(2) \Rightarrow f(0) > f(2)$

$\Rightarrow$ f(x) is not increasing function

Let f(x) is decreasing function f(f(f(x))) is decreasing function $f(f(f(0))) > f(f(f(2)))$

$\Rightarrow 2 - f(0) > 2 - f(2) \Rightarrow f(0) < f(2)$

Hence, f(x) is not decreasing function

Since, f(x) is differentiable and does not have local maximum or minimum

$\Rightarrow$ f(x) is a constant function i.e. $f(x) = 1 \Rightarrow \int_0^2 f(x)dx = \int_0^2 dx = 2$

40. A, D

Sol. Since, $f'(x) > 0 \Rightarrow f(x)$ is increasing function

$\Rightarrow f(0) = 0, f(1) = 1$

Let $g(x) = \frac{f(x)}{x}$, $x \in (0, 1] \Rightarrow g'(x) = \frac{xf'(x) - f(x)}{x^2}$

Let $p(x) = xf'(x) - f(x)$; $x \in [0, 1]$

$\Rightarrow p'(x) = xf''(x) + f'(x) - f'(x) < 0 \ \forall x \in (0, 1]$

$\Rightarrow p(x) < p(0) = -f(0) = 0 \Rightarrow g'(x) < 0 \ \forall x \in (0, 1]$

$\Rightarrow \frac{f(x)}{x} \ge \frac{f(1)}{1} \Rightarrow \frac{f(x)}{x} \ge 1 \Rightarrow f(x) \ge x \Rightarrow x \ge f^{-1}(x)$

41. B, C

Sol. $a_n = 2a_{n-1} - 1$, let $a_n = 2^n t_n$

$\Rightarrow 2^n t_n = 2^n t_{n-1} - 1 \Rightarrow t_n - t_{n-1} = -2^{-n}$

$\Rightarrow \sum_{n=2}^{n}(t_n - t_{n-1}) = \sum_{n=2}^{n}(-2^{-n}) \Rightarrow t_n - t_1 = -\frac{1}{2} + \frac{1}{2^n}$ where $a_1 = 2t_1 = 3 \Rightarrow t_1 = \frac{3}{2}$

$\Rightarrow t_n = 1 + \frac{1}{2^n} \Rightarrow 2^n t_n = 2^n + 1 \Rightarrow a_n = 2^n + 1$

42. B

Sol. Let $B = [b_{ij}]_{n\times n}$ such that $b_{ij} = 1 \ \forall$ i, j

$\Rightarrow B^2 = nB$ and $A = (n + 1)I - B$

Let C be $n \times n$ matrix such that $C = A^{-1}$

$\Rightarrow C = aI + bB \Rightarrow AC = I \Rightarrow ((n + 1)I - B)(aI + bB) = I$

$\Rightarrow a(n + 1)I - aB + (n + 1)bB - bB^2 = I$

$\Rightarrow (a(n + 1) - 1)I = aB - (n + a)bB + bnB = (a - (n + 1)b + nb)B = (a - b)B$

$\Rightarrow an + a - 1 = a - b$ and $a - b = 0$

$a = \frac{1}{n+1} = b \Rightarrow C = \frac{1}{n+1}(I + B)$

43. C

Sol. $n(S) = {}^9C_3 \cdot {}^8C_3$

$$n(E) = \frac{1}{2}\left({}^8C_3{}^8C_3 + {}^8C_3\right) + {}^8C_2{}^8C_3$$

$$= \frac{1}{2}\cdot {}^8C_3\left({}^8C_3 + 1 + 2\cdot{}^8C_3\right) = 28(56 + 1 + 56) = 28(113)$$

$$\Rightarrow p(E) = \frac{28(113)}{56\times 84} = \frac{113}{168}$$

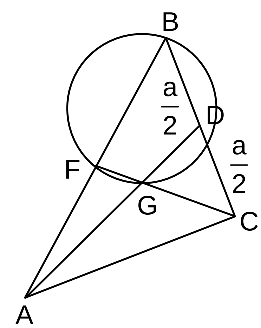
44. A, B, D

Sol. Distance of A″ from plane $2x - 2y + z = 0$ should be 1 unit

45. C

46. A

Sol. **(for Q. 45.-46)**



We know that CD·CB = CG·CF

$$\frac{a^2}{2} = \frac{2}{3}(CF)^2 \Rightarrow 3a^2 = 4\left(\frac{2a^2 + 2b^2 - c^2}{4}\right)$$

$\Rightarrow a^2 = 2b^2 - c^2$

$\Rightarrow c^2 = (2b^2 - a^2)$ ..... (1)

and $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$

For $C \in \left(\dfrac{2\pi}{3}, \pi\right)$, $-1 < \cos\dfrac{2\pi}{3} < -\dfrac{1}{2}$

$$-1 < \frac{a^2 + b^2 - (2b^2 - a^2)}{2ab} < -\frac{1}{2} \Rightarrow -2ab < 2a^2 - b^2 < -ab$$

$\Rightarrow 2a^2 + 2ab - b^2 > 0$ and $2a^2 + ab - b^2 < 0$

$$\left(\frac{a}{b} + \left(\frac{1+\sqrt{3}}{2}\right)\right)\left(\frac{a}{b} - \left(\frac{-1+\sqrt{3}}{2}\right)\right) > 0 \text{ and } (2a - b)(a + b) < 0$$

$$\frac{\sqrt{3}-1}{2} < \frac{a}{b} < \frac{1}{2} \text{ when } c = \frac{2\pi}{3},\ \frac{a^2 + b^2 - c^2}{2ab} = -\frac{1}{2}$$

$$a^2 + b^2 - (2b^2 - a^2) = -ab \Rightarrow 2a^2 - b^2 + ab = 0 \Rightarrow \frac{a}{b} = \frac{1}{2},\ c^2 = 7a^2$$

$\Rightarrow a^2 = 7,\ b = 2\sqrt{7}$

47. B

Sol. $n(S) = D_5 \cdot 7!$; $n(E) = D_5 \cdot D_7$

$$\Rightarrow P(E) = \frac{D_5\cdot D_7}{D_5\cdot 7!} = \frac{7!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!}\right)}{7!}$$

$$= \frac{2520 - 840 + 210 - 42 + 7 - 1}{7!} = \frac{1854}{7!}$$

48. C

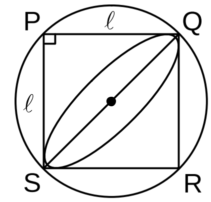
Sol. $n(S) = 7!\,7!$; $n(E) = 1\cdot D_7$

$$\Rightarrow P(E) = \frac{D_7}{7!\,7!} = \frac{1854}{(7!)^2}$$

Section – B

49. 194

Sol. PQRS is a square



$\Rightarrow \ell^2 = A \Rightarrow \ell = \sqrt{A}$ where $\sqrt{2}\ell = 2\sqrt{a^2+b^2}$

$\Rightarrow \sqrt{A} = \sqrt{2(a^2+b^2)} \Rightarrow a^2+b^2 = \dfrac{A}{2}$ and a – b = r

$\Rightarrow a^2 + b^2 - 2ab = r^2 \Rightarrow 2ab = \left(\dfrac{A}{2} - r^2\right)$

$(a+b)^2 = \dfrac{A}{2} + \left(\dfrac{A}{2} - r^2\right) = (A - r^2) \Rightarrow a+b = \sqrt{A-r^2}$

$\Rightarrow 2a = r + \sqrt{A-r^2}$, $2b = \sqrt{A-r^2} - r$

$$e^2 = 1 - \left(\frac{b^2}{a^2}\right) \Rightarrow \frac{1}{4} = 1 - \left(\frac{\sqrt{A-r^2}-r}{\sqrt{A-r^2}+r}\right)^2 \Rightarrow \frac{r^2\left(\sqrt{\frac{A}{r^2}-1}-1\right)^2}{r^2\left(\sqrt{\frac{A}{r^2}-1}+1\right)^2} = \frac{3}{4} \Rightarrow \frac{\sqrt{\frac{A}{r^2}-1}+1}{\sqrt{\frac{A}{r^2}-1}-1} = \frac{2}{\sqrt{3}}$$

$$\sqrt{3}\sqrt{\frac{A}{r^2}-1} + \sqrt{3} = 2\sqrt{\frac{A}{r^2}-1} - 2\ ;\ (2-\sqrt{3})\sqrt{\frac{A}{r^2}-1} = (2+\sqrt{3}) \Rightarrow \frac{A}{r^2} - 1 = (2+\sqrt{3})^4$$

$\therefore \left[\dfrac{A}{r^2}\right] = 194$

50. 0

Sol. $\cos z = \dfrac{e^{iz}+e^{-iz}}{2}$, $\sin z = \dfrac{e^{iz}-e^{-iz}}{2i}$

$\Rightarrow \dfrac{e^{iz}+e^{-iz}}{2} = \dfrac{e^{iz}-e^{-iz}}{2} \Rightarrow e^{-iz} = 0$

Let z = x + iy $\Rightarrow e^{-i(x+iy)} = 0$

$\Rightarrow e^{y-ix} = 0 \Rightarrow e^{y}(\cos x - i\sin x) = 0$

$\Rightarrow$ tan x = i $\Rightarrow x \notin R \Rightarrow$ No real solution exists

51. 12

Sol. a, b, c $\in$ N

The remainder when a^2, b^2, c^2 is divided by 13 is {0, 1, 3, 4, 9, 10, 12}

Let E be the event that $a^2 + b^2 + c^2 = 13k$

$\Rightarrow P(E) = \dfrac{1+3\times1\times2\times2\times3!+2\times2\times2\times3!+2\times2\times2\times3!}{13^3} = \dfrac{169}{13^3} = \dfrac{1}{13}$

$\Rightarrow$ 13 – 1 = 12

Section – C

52. 2.00

53. 4.00

Sol. $\alpha = \beta = 2$

54. 00.75

55. 00.25

Sol. **(for Q. 54.-55)**

We know that $\displaystyle\int_0^x \left(1+t+t^2+t^3+.....\infty\right)dt = \int_0^x \left(\frac{1}{1-x}\right)dt$

$$\Rightarrow x+\frac{x^2}{2}+\frac{x^3}{3}+\ldots..\infty=-\ln(1-x)$$

$$\Rightarrow 1+\frac{x}{2}+\frac{x^2}{3}+\ldots..\infty=-\frac{\ln(1-x)}{x}$$

$$\Rightarrow \int_0^1\left(1+\frac{x}{2}+\frac{x^2}{3}+\ldots..\infty\right)dx=-\int_0^1\frac{\ln(1-x)}{x}dx$$

$$\Rightarrow -\int_0^1\frac{\ln x}{1-x}=a \Rightarrow \int_0^1 f(x)dx=-a \Rightarrow I_1=-a$$

$$I_2=\int_0^1\left(\frac{x\ln x}{1-x^2}\right)dx\text{ , let } x^2=t \Rightarrow 2x\,dx=dt$$

$$I_2=\frac{1}{2}\int_0^1\left(\frac{\ln\sqrt{t}}{1-t}\right)dt=\frac{1}{4}\int_0^1\left(\frac{\ln t}{1-t}\right)dt=\frac{1}{4}I_1 \Rightarrow I_1-I_2=\frac{3}{4}I_1=-\frac{3}{4}a$$

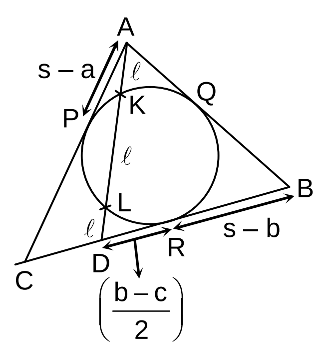
$$\Rightarrow \frac{I_1-I_2}{a}=-\frac{3}{4}\text{ and }\lim_{x\to\infty}\sum_{r=1}^{n}\frac{r\ln\left(\frac{r}{n}\right)}{n^2\left(1-\frac{r^2}{n^2}\right)}=\lim_{x\to\infty}\sum_{r=1}^{n}\frac{1}{x}\cdot\frac{r}{x}\cdot\frac{\ln\left(\frac{r}{n}\right)}{1-\left(\frac{r}{n}\right)^2}$$

$$=\int_0^1\frac{x\,dx}{1-x^2}=\frac{1}{4}(-a)=-\frac{a}{4} \Rightarrow \lambda=-\frac{1}{4}$$

56. 14.48

57. 01.60

Sol. **(for Q. 56.-57)**



$AP = DR = \sqrt{\ell(2\ell)}$

$\Rightarrow s - a = \frac{b-c}{2}$

$\Rightarrow \frac{b+c-a}{2}=\frac{b-c}{2}$

$\Rightarrow a = 2c$

Also, $2\ell^2=\left(\frac{b-c}{2}\right)^2$ ..... (1)

And median AD $\Rightarrow 3\ell=\frac{1}{2}\sqrt{2b^2+2c^2-a^2}$

$\Rightarrow 18(2\ell^2)=2b^2+2c^2-a^2$; $18\left(\frac{b-c}{2}\right)^2=2b^2+2c^2-4c^2$ (Using equation (1))

$\Rightarrow \frac{9}{2}(b-c)^2=2(b^2-c^2) \Rightarrow 9(b + c) = 4(b – c) \Rightarrow 5b = 13c$

$\Rightarrow \frac{a}{2}=\frac{5b}{13}=c \Rightarrow \frac{a}{10}=\frac{b}{13}=\frac{c}{5}=t \Rightarrow a = 10t, b = 13t, c = 5t, 2s = 28t = 28 \Rightarrow t = 1$

$\Rightarrow a = 10, b = 13, c = 5 \Rightarrow \Delta=\sqrt{s(s-a)(s-b)(s-c)}=6\sqrt{14}$

$\Rightarrow r=\frac{\Delta}{s}=\frac{6\sqrt{14}}{14}=\frac{3}{7}\sqrt{14}$ also, $\Delta=\frac{abc}{4R} \Rightarrow 2R=\frac{abc}{2\Delta}=\frac{10\cdot 13\cdot 5}{2\cdot 6\sqrt{14}}=\frac{325}{6\sqrt{14}}$